



Microwaves are bad for the Earth

Microwaves in EU emit as much carbon dioxide as 7 million cars, new research shows.
<http://digit.in/mwco2>



China is collecting environment tax

In a bid to tackle increasing pollution, China has started taxing companies for producing waste.
<http://digit.in/taxcne>

in the tundra, is a trapped layer of permafrost - a frozen layer of ice and soil. Trapped within the permafrost is methane produced by bacteria, as well as significant amounts of carbon dioxide. The problem is that the permafrost is no longer frozen, as the temperatures are dropping below zero. The explanation most favored by researchers, is global warming, brought about by increased human industrial activity. Another explanation is that heat is seeping up through the cracks in the rocks, because of the tectonic activity in the region. A few days of sudden increase in heat can also cause disruptions in the permafrost layer.

AN ENVIRONMENTAL TIME BOMB

Once the permafrost melts, the trapped methane is released below the ground. This causes the ground to swell up, creating mounds known pingos. The mounds wobble, similar to jelly. With a build up of sufficient pressure, the mounds can explode, leaving a crater behind, and releasing the methane into the atmosphere. While most of the craters are believed to have been formed due to pingo explosions, there are collapsed craters that have appeared where pingos were not present.

To the north of the Yamal peninsula, is a tiny island called Bely, with a number of pingos. The ground wobbles, as if it were a giant trampoline. Gases leak out into the atmosphere from beneath the ground. In an expedition to Bely island, researchers measured exactly how toxic the gases released by the melting permafrost were. Measurements in 2016 indicated that the leaking air had 20 times as much carbon dioxide and about 200 times as much methane as the regular air in the atmosphere. Follow up measurements in 2017 revealed something even more startling. The level of carbon dioxide was 25 times more, but

the level of methane was in fact closer to 1000 times more.

Both carbon dioxide and methane are greenhouse gases that trap heat and contribute to global warming. While there is a lot of focus on reducing the amount of carbon dioxide pumped into the atmosphere, there is less attention on methane because of its low percentage in emissions. However, methane is actually between 30 to 80 times more potent at trapping heat than carbon dioxide. Large amounts of methane and carbon dioxide emissions would form a positive feedback loop. More greenhouse gases means more heat in the atmosphere, and more heat means



The Yamal Crater.

more greenhouse gases get emitted. The mounds exist in the shallow waters near the Siberian shelf as well. A vessel almost capsized in 1995, because of a bubble of methane that it accidentally drilled into. Researchers estimate that there are over 7,000 gas bubbles in Siberia that are expected to explode. From the Arctic seabed, the methane is leaking out in bubbles from areas known as taliks, which are releasing as much methane into the atmosphere as the land.

In 2013, an international team of researchers went into the caves along the permafrost frontier to better understand how changing climate conditions affects the permafrost. These caves are located in the layer of the ground, where the soil and ice is permanently frozen. The stalactites and stalagmites in these caves provide a record of over 500,000 years of

weather conditions. The formations only grow when there is flowing water, which means when the temperature is warmer. The largest periods of growth were observed over 400,000 years ago, when the atmosphere was about 1.5 degrees warmer. This means that an increase of 1.5 degrees of global temperature, is the tipping point, at which large amounts of permafrost will begin to melt. Permafrost covers about 24 percent of the land area in the northern hemisphere, and can potentially release gigatonnes of greenhouse gases into the atmosphere.

And it is not just greenhouse gases, there are other ancient terrors locked within the permafrost as well. In 2016, there was a sudden heat wave where temperatures soared to 35 degrees Celsius in the Yamalo-Nenets Autonomous Okrug. One of the things exposed by the melting permafrost, was a reindeer carcass infected with anthrax. First, the local reindeer were infected, with over 2,300 dying because of the disease. The sickness jumped to humans from the reindeer, which resulted in several confirmed infections, and the death of a twelve year old boy. It was the first breakout of anthrax in the region since 1941.

INDUSTRIALISATION TO THE RESCUE

There are massive energy projects planned in the Yamal region, considering the area has one of the biggest gas deposits in the world. Professor Anthony Hopkins from the Australian National University has a brilliant solution. Turns out, that the industry has the technologies necessary to harvest the gases trapped within the permafrost. The same techniques used to harvest the gas can also be used to harvest the methane trapped within the ground. This methane could be burnt up to form carbon dioxide, which is actually more preferable considering how toxic methane is. However, successfully getting the gas industry to do this would require some kind of subsidies from the government, or external funding. 